

# OpenDC-LCA: from heterogeneous evidence to bounded cooling decisions

One calculation spine keeps the functional unit, boundary, transformation, uncertainty and claim status visible.

## DECISION QUESTION

Which cooling architecture is preferred, where can the ranking change, and what evidence would resolve it?

### 1 Evidence with declared role

#### Released lifecycle model

Microsoft/WSP component results  
Role: arithmetic reconstruction

#### Public scenario evidence

eGRID · Boavizta · ÖKOBAUDAT  
Role: bounded screening stress

#### Exchange-ready inventories

USLCI · GLAD · openLCA/Brightway  
Role: registered, not silently combined

#### Primary engineering evidence

Performance maps · BOM · failures  
Role: required for decision-grade claims

### 2 Harmonization and claim gate

1

#### Define equivalence

Useful computation · lifetime · reliability

2

#### Align what can be aligned

Unit · GWP basis · year · geography

3

#### Expose transformation limits

Interpolation · extrapolation · boundary gap

4

#### Propagate and classify evidence

Joint stress · provenance · review status

SCREENING ≠ COMPARATIVE ASSERTION

### 3 Decision outputs

#### Reproducibility

24 / 24

released totals reconciled  
source IDs + equations + digests

#### Conditional decision boundary

96.7

kg CO<sub>2</sub>e/MWh  
cold plate / one-phase crossover

#### Robustness and next evidence

29.8%

outside released anchors  
99 → 55%<sup>P</sup> first-rank stress frequency

Output: crossover + stress envelope + acquisition plan

● reconstructed ● screening transformation ● registered / unlinked ● primary / decision-grade

Every numerical claim retains its evidence class.